



Started on	Thursday, 24 September 2026, 10:24 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:34 AM
Time taken	9 mins 25 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6      int arr[n];
7      for(int i=0;i<n;i++)
8      {
9          scanf("%d",&arr[i]);
10     }
11     for(int i=0;i<n;i++)
12     {
13         for(int j=i+1;j<n;j++)
14         {
15             if(arr[i]==arr[j])
16             {
17                 printf("%d", arr[i]);
18             }
19         }
20     }
21 }
22 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 24 September 2026, 10:34 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:58 AM
Time taken	23 mins 50 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     int arr[n];
7     for(int i=0;i<n;i++)
8     {
9         scanf("%d",&arr[i]);
10    }
11    int slow=arr[0];
12    int fast=arr[0];
13
14    do
15    {
16        slow=arr[slow];
17        fast=arr[arr[fast]];
18    }while(slow!=fast);
19    slow=arr[0];
20    while(slow!=fast)
21    {
22        slow=arr[slow];
23        fast=arr[fast];
24    }
25    printf("%d",slow);
26    return 0;
27 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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Harini P 251801087-2025-AIDS ▾

H2**Started on** Thursday, 24 September 2026, 10:58 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:12 AM**Time taken** 14 mins 15 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int t;
6     scanf("%d", &t);
7
8     while (t--)
9     {
10         int n;
11         scanf("%d", &n);
12
13         int arr[n];
14
```

```
15     for (int i = 0; i < n; i++)
16     {
17         scanf("%d", &arr[i]);
18     }
19
20     int m;
21     scanf("%d", &m);
22
23     int brr[m];
24
25     for (int i = 0; i < m; i++)
26     {
27         scanf("%d", &brr[i]);
28     }
29
30     for (int i = 0; i < n; i++)
31     {
32         for (int j = 0; j < m; j++)
33         {
34             if (arr[i] == brr[j])
35             {
36                 printf("%d ", arr[i]);
37                 break;
38             }
39         }
40     }
41
42     printf("\n");
43 }
44
45 return 0;
46 }
47
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Started on** Thursday, 24 September 2026, 11:12 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:19 AM**Time taken** 6 mins 24 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int t;
6     scanf("%d", &t);
7
8     while (t--)
9     {
10         int m, n;
11
12         scanf("%d", &m);
13
14         int a[m];
```

```
15     for (int i = 0; i < m; i++)
16         scanf("%d", &a[i]);
17
18     scanf("%d", &n);
19
20     int b[n];
21     for (int i = 0; i < n; i++)
22         scanf("%d", &b[i]);
23
24     int i = 0, j = 0;
25
26     while (i < m && j < n)
27     {
28         if (a[i] < b[j])
29             i++;
30         else if (a[i] > b[j])
31             j++;
32         else
33         {
34             printf("%d ", a[i]);
35             i++;
36             j++;
37         }
38     }
39
40     printf("\n");
41 }
42
43 return 0;
44 }
45
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 24 September 2026, 11:19 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:26 AM
Time taken	6 mins 26 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, k;
6
7      scanf("%d", &n);
8
9      int a[n];
10
11     for (int i = 0; i < n; i++)
12     {
13         scanf("%d", &a[i]);
14     }
15
16     scanf("%d", &k);
17
18     for (int i = 0; i < n; i++)
19     {
20         for (int j = i + 1; j < n; j++)
21         {
22             if (a[j] - a[i] == k)
23             {
24                 printf("1");
25                 return 0;
26             }
27         }
28     }
29     printf("0");
30     return 0;
31 }
32

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 24 September 2026, 11:26 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:34 AM
Time taken	8 mins 14 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, k;
6      scanf("%d", &n);
7      int a[n];
8      for (int i = 0; i < n; i++)
9      {
10         scanf("%d", &a[i]);
11     }
12     scanf("%d", &k);
13     int i = 0, j = 1;
14     while (i < n && j < n)
15     {
16         if (i == j)
17         {
18             j++;
19         }
20         else if (a[j] - a[i] < k)
21         {
22             j++;
23         }
24         else if (a[j] - a[i] > k)
25         {
26             i++;
27         }
28         else
29         {
30             printf("1");
31             return 0;
32         }

```

```
33 |     }  
34 |     printf("0");  
35 |     return 0;  
36 | }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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